

Customer Retention Analysis Using Behavioral and Product Engagement in Banking

Abstract:

Customer churn remains a critical challenge in the banking sector. This study analyzes retention using engagement metrics and machine learning. Results show engagement and product usage strongly impact retention.

Introduction:

Retention is more cost-effective than acquisition. This study focuses on behavioral engagement rather than demographics.

Problem Statement:

Banks lack insights into behavioral churn drivers and rely too much on financial indicators.

Objectives:

- Analyze engagement vs churn
- Study product impact
- Identify high-risk customers

Methodology:

Data preprocessing, feature engineering (Engagement Score), and Random Forest model were used.

Results:

- Active users churn less
- Multi-product users are more loyal
- High balance $\neq$ high loyalty

Conclusion:

Engagement-driven strategies improve retention more than financial targeting.

Future Scope:

Real-time analytics, AI models, and personalization.